

ES Lab3 optional problem A

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./examples/ble_scan/ble_scan

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deep90@dog:~/gattlib/build$ sudo ./examples/ble_scan/ble_scan
Discovered 70:06:A9:A2:A9:E6
Discovered 7A:78:23:08:F1:95 - 'Blank'
Discovered 7D:44:02:9A:74:09 - '534AG7-Find My'
Discovered 69:51:2E:AD:E6:EC
Discovered 3F:BD:CF:B9:50:F6
Discovered 71:1F:4B:CA:8F:BB
Discovered 78:AC:00:F9:38:89
Discovered 13:5B:5C:EF:7B:90
Discovered A4:C1:38:60:60:76 - 'LYWSD03MMC'
Discovered EC:2C:1A:AB:FF:20
Discovered FC:F3:43:00:23:5A
Discovered 2B:DE:65:6E:73:92
Scan completed
-----START 70:06:A9:A2:A9:E6 -----
gattlib ble_scan: Device connected error (device:/org/bluez/hci0/dev_70_06_A9_A2_A9_E6): GDBus.Error:org.bluez.Error.Failed: le-connection-abort-by-local
gattlib ble_scan: Gattlib connection not initialized.
gattlib ble_scan: Fail to discover primary services.
gattlib ble_scan: Cannot disconnect - connection parameter is not valid.
gattlib ble_scan: Failed to connect to the bluetooth device '70:06:A9:A2:A9:E6'
-----DONE 70:06:A9:A2:A9:E6 -----
-----START 7A:78:23:08:F1:95 -----
-----DONE 7A:78:23:08:F1:95 -----
-----START 7D:44:02:9A:74:09 -----
gattlib ble_scan: Device connected error (device:/org/bluez/hci0/dev_7D_44_02_9A_74_09): GDBus.Error:org.bluez.Error.Failed: le-connection-abort-by-local
gattlib ble_scan: Gattlib connection not initialized.
gattlib ble_scan: Fail to discover primary services.
gattlib ble_scan: Cannot disconnect - connection parameter is not valid.
gattlib ble_scan: Failed to connect to the bluetooth device '7D:44:02:9A:74:09'
-----DONE 7D:44:02:9A:74:09 -----
-----START 69:51:2E:AD:E6:EC -----
gattlib ble_scan: Device connected error (device:/org/bluez/hci0/dev_69_51_2E_AD_E6_EC): GDBus.Error:org.bluez.Error.Failed: le-connection-abort-by-local
gattlib ble_scan: Gattlib connection not initialized.
gattlib ble_scan: Fail to discover primary services.
gattlib ble_scan: Cannot disconnect - connection parameter is not valid.
gattlib ble_scan: Failed to connect to the bluetooth device '69:51:2E:AD:E6:EC'
-----DONE 69:51:2E:AD:E6:EC -----
-----START 3F:BD:CF:B9:50:F6 -----
-----DONE 3F:BD:CF:B9:50:F6 -----
-----START 71:1F:4B:CA:8F:BB -----
gattlib ble_scan: Device connected error (device:/org/bluez/hci0/dev_71_1F_4B_CA_8F_BB): GDBus.Error:org.bluez.Error.Failed: le-connection-abort-by-local
gattlib ble_scan: Gattlib connection not initialized.
gattlib ble_scan: Fail to discover primary services.
gattlib ble_scan: Cannot disconnect - connection parameter is not valid.
gattlib ble_scan: Failed to connect to the bluetooth device '71:1F:4B:CA:8F:BB'
-----DONE 71:1F:4B:CA:8F:BB -----
-----START 78:AC:00:F9:38:89 -----
gattlib ble_scan: Device connected error (device:/org/bluez/hci0/dev_78_AC_00_F9_38_89): Timeout was reached
gattlib ble_scan: Gattlib connection not initialized.
gattlib ble_scan: Fail to discover primary services.
gattlib ble_scan: Cannot disconnect - connection parameter is not valid.
gattlib ble_scan: Failed to connect to the bluetooth device '78:AC:00:F9:38:89'
-----DONE 78:AC:00:F9:38:89 -----
-----START 13:5B:5C:EF:7B:90 -----
gattlib ble_scan: Device '13:5B:5C:EF:7B:90' cannot be found (97, 41)
gattlib ble_scan: Gattlib connection not initialized.
gattlib ble_scan: Fail to discover primary services.
gattlib ble_scan: Cannot disconnect - connection parameter is not valid.
gattlib ble_scan: Failed to connect to the bluetooth device '13:5B:5C:EF:7B:90'
-----DONE 13:5B:5C:EF:7B:90 -----
-----START A4:C1:38:60:60:76 -----
service[0] start_handle:2d end_handle:36 uuid:89d3502b-0f36-433a-8ef4-c502ad55f8dc
service[1] start_handle:01 end_handle:04 uuid:0x1800
service[2] start_handle:06 end_handle:07 uuid:0x1801
service[3] start_handle:0a end_handle:0d uuid:0x180a
service[4] start_handle:0f end_handle:10 uuid:d0611e78-bbb4-4591-a5f8-487910ae4366
service[5] start_handle:14 end_handle:15 uuid:9fa480e0-4967-4542-9390-d343dc5d04ae
service[6] start_handle:19 end_handle:1a uuid:0x180f
service[7] start_handle:23 end_handle:2a uuid:7905f431-b5ce-4e99-a40f-4b1e22d00d0
service[8] start_handle:39 end_handle:3a uuid:0xffff
service[9] start_handle:1d end_handle:21 uuid:0x1805
-----DONE A4:C1:38:60:60:76 -----
-----START EC:2C:1A:AB:FF:20 -----
characteristic[0] properties:18 value_handle:0032 uuid:2f7cabce-880d-411f-9a0c-bb92ba96c102
characteristic[1] properties:0a value_handle:0036 uuid:c6b2f38c-23ab-46d8-a6ab-a3a870bbd5d7
characteristic[2] properties:18 value_handle:002e uuid:9b3c81d8-57b1-4a8a-b8df-0e56f7ca51c2
characteristic[3] properties:02 value_handle:0002 uuid:0x2a00
characteristic[4] properties:02 value_handle:0004 uuid:0x2a01
characteristic[5] properties:29 value_handle:0007 uuid:0x2a05
characteristic[6] properties:02 value_handle:000d uuid:0x2a24
characteristic[7] properties:02 value_handle:000b uuid:0x2a29
characteristic[8] properties:18 value_handle:0010 uuid:8667556c-9a37-4c91-84ed-54ee27d90049
characteristic[9] properties:18 value_handle:0015 uuid:af0badb1-5b99-43cd-917a-a77bc549e3cc
characteristic[10] properties:12 value_handle:001a uuid:0x2a19
characteristic[11] properties:08 value_handle:0024 uuid:69d1d8f3-45e1-49a8-9021-9bbdfdaad9d9
characteristic[12] properties:10 value_handle:002a uuid:22eac6e9-24d6-4bb5-be44-b36ace7c7bfb
characteristic[13] properties:10 value_handle:0027 uuid:9fbf120d-6301-42d9-8c58-25e699a21dbd
characteristic[14] properties:32 value_handle:003a uuid:0xffff1
characteristic[15] properties:12 value_handle:001e uuid:0x2a2b
characteristic[16] properties:02 value_handle:0021 uuid:0x2a0f
-----DONE EC:2C:1A:AB:FF:20 -----
-----START FC:F3:43:00:23:5A -----
gattlib ble_scan: Device connected error (device:/org/bluez/hci0/dev_FC_F3_43_00_23_5A): GDBus.Error:org.bluez.Error.Failed: le-connection-abort-by-local
gattlib ble_scan: Gattlib connection not initialized.
gattlib ble_scan: Fail to discover primary services.
gattlib ble_scan: Cannot disconnect - connection parameter is not valid.
gattlib ble_scan: Failed to connect to the bluetooth device 'FC:F3:43:00:23:5A'
-----DONE FC:F3:43:00:23:5A -----
-----START 2B:DE:65:6E:73:92 -----
service[0] start_handle:82 end_handle:83 uuid:0x1855
service[1] start_handle:14 end_handle:19 uuid:0x1800
service[2] start_handle:5a end_handle:7c uuid:0x184c
service[3] start_handle:28 end_handle:4c uuid:0x1849
service[4] start_handle:0a end_handle:0e uuid:0xfcf3
service[5] start_handle:03 end_handle:07 uuid:594a34fc-31db-11ea-978f-2e720ce88125
service[6] start_handle:01 end_handle:08 uuid:0x1801
service[7] start_handle:00 end_handle:91 uuid:e73e0001-ef1b-4e74-8291-2e4f3164f3b5
service[8] start_handle:a0 end_handle:a1 uuid:0xfcfc1
characteristic[0] properties:02 value_handle:0083 uuid:0x2b51
characteristic[1] properties:02 value_handle:0019 uuid:0x2aa6
characteristic[2] properties:02 value_handle:0017 uuid:0x2a01
characteristic[3] properties:02 value_handle:0015 uuid:0x2a00
characteristic[4] properties:02 value_handle:005e uuid:0x2bb4
characteristic[5] properties:12 value_handle:0079 uuid:0x2bc1
characteristic[6] properties:12 value_handle:005b uuid:0x2bb3
characteristic[7] properties:10 value_handle:0076 uuid:0x2bc0
characteristic[8] properties:12 value_handle:007c uuid:0x2bc2
characteristic[9] properties:02 value_handle:0074 uuid:0x2bbf
characteristic[10] properties:1c value_handle:0071 uuid:0x2bbe
characteristic[11] properties:02 value_handle:0069 uuid:0x2bba
characteristic[12] properties:12 value_handle:000e uuid:0x2bbd
characteristic[13] properties:12 value_handle:0066 uuid:0x2bb9
characteristic[14] properties:12 value_handle:006b uuid:0x2bbb
characteristic[15] properties:12 value_handle:0063 uuid:0x2bb6
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characteristic[16] properties:12 value_handle:0060 uuid:0x2bb5
characteristic[17] properties:10 value_handle:002c uuid:0x2b96
characteristic[18] properties:1c value_handle:0046 uuid:0x2ba4
characteristic[19] properties:02 value_handle:004c uuid:0x2bba
characteristic[20] properties:12 value_handle:0043 uuid:0x2ba3
characteristic[21] properties:02 value_handle:0041 uuid:0x2ba2
characteristic[22] properties:1e value_handle:0038 uuid:0x2b9a
characteristic[23] properties:1e value_handle:003e uuid:0x2ba1
characteristic[24] properties:1e value_handle:0035 uuid:0x2b99
characteristic[25] properties:12 value_handle:003b uuid:0x2b9b
characteristic[26] properties:12 value_handle:0032 uuid:0x2b98
characteristic[27] properties:12 value_handle:0029 uuid:0x2b93
characteristic[28] properties:12 value_handle:002f uuid:0x2b97
characteristic[29] properties:12 value_handle:0049 uuid:0x2ba5
characteristic[30] properties:08 value_handle:009e uuid:00000100-0004-1000-8000-001a11000101
characteristic[31] properties:20 value_handle:009b uuid:00000100-0004-1000-8000-001a11000102
characteristic[32] properties:0c value_handle:0097 uuid:594a3010-31db-11ea-978f-2e728ce88125
characteristic[33] properties:28 value_handle:0094 uuid:594a36e6-31db-11ea-978f-2e728ce88125
characteristic[34] properties:02 value_handle:0004 uuid:0x2b3a
characteristic[35] properties:20 value_handle:0002 uuid:0x2a05
characteristic[36] properties:02 value_handle:0008 uuid:0x2b2a
characteristic[37] properties:0a value_handle:0006 uuid:0x2b29
characteristic[38] properties:02 value_handle:0091 uuid:e73e0002-ef1b-4e74-8201-2e4f3164f3b5
characteristic[39] properties:20 value_handle:00a1 uuid:fe2c1237-8366-4814-8eb0-01de32100bea
-----DONE 78:DE:05:0E:73:92 -----
gattlib ble scan: Adapter cannot be closed as some devices are not disconnected

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./examples/discover/discover 78:70:9B:A0:45:BD

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dog50301@dog:~/gattlib/build$ sudo ./examples/discover/discover 78:70:9B:A0:45:BD
gattlib discover: Found bluetooth device '78:70:9B:A0:45:BD'
gattlib discover: Connected to bluetooth device '78:70:9B:A0:45:BD'
gattlib discover: service[0] start_handle:22 end_handle:26 uuid:0x1805
gattlib discover: service[1] start_handle:32 end_handle:3b uuid:89d3502b-0f36-433a-8ef4-c502ad55f8dc
gattlib discover: service[2] start_handle:0f end_handle:10 uuid:d0611e78-bbb4-4591-a5f8-487910ae4366
gattlib discover: service[3] start_handle:06 end_handle:07 uuid:0x1801
gattlib discover: service[4] start_handle:1e end_handle:1f uuid:0x180f
gattlib discover: service[5] start_handle:28 end_handle:2f uuid:7905f431-b5ce-4e99-a40f-4b1e122d00d0
gattlib discover: service[6] start_handle:0a end_handle:0d uuid:0x180a
gattlib discover: service[7] start_handle:14 end_handle:15 uuid:9fa480e0-4967-4542-9390-d343dc5d04ae
gattlib discover: service[8] start_handle:3e end_handle:3f uuid:0xffff0
gattlib discover: service[9] start_handle:01 end_handle:04 uuid:0x1800
gattlib discover: characteristic[0] properties:02 value_handle:0026 uuid:0x2a0f
gattlib discover: characteristic[1] properties:12 value_handle:0023 uuid:0x2a2b
gattlib discover: characteristic[2] properties:0a value_handle:003b uuid:c6b2f38c-23ab-46d8-a6ab-a3a870bbd5d7
gattlib discover: characteristic[3] properties:18 value_handle:0033 uuid:9b3c81d8-57b1-4a8a-b8df-0e56f7ca51c2
gattlib discover: characteristic[4] properties:18 value_handle:0037 uuid:2f7cabce-808d-411f-9a0c-bb92ba96c102
gattlib discover: characteristic[5] properties:18 value_handle:0010 uuid:8667556c-9a37-4c91-84ed-54ee27d90049
gattlib discover: characteristic[6] properties:20 value_handle:0007 uuid:0x2a05
gattlib discover: characteristic[7] properties:12 value_handle:001f uuid:0x2a19
gattlib discover: characteristic[8] properties:08 value_handle:0029 uuid:69d1d8f3-45e1-49a8-9821-9bbdfdaad9d9
gattlib discover: characteristic[9] properties:10 value_handle:002f uuid:22eac6e9-24d6-4bb5-be44-b36ace7c7bfb
gattlib discover: characteristic[10] properties:10 value_handle:002c uuid:9fbf120d-6301-42d9-8c58-25e699a21ddb
gattlib discover: characteristic[11] properties:02 value_handle:000d uuid:0x2a24
gattlib discover: characteristic[12] properties:02 value_handle:000b uuid:0x2a29
gattlib discover: characteristic[13] properties:18 value_handle:0015 uuid:af0badb1-5b99-43cd-917a-a77bc549e3cc
gattlib discover: characteristic[14] properties:3a value_handle:003f uuid:0xffff1
gattlib discover: characteristic[15] properties:02 value_handle:0004 uuid:0x2a01
gattlib discover: characteristic[16] properties:02 value_handle:0002 uuid:0x2a00

```

sudo ./examples/read_write/read_write 78:70:9B:A0:45:BD read 0000fff1-0000-1000-8000-00805f9b34fb

```

dog50301@dog:~/gattlib/build$ sudo ./examples/read_write/read_write 78:70:9B:A0:45:BD read 0000fff1-0000-1000-8000-00805f9b34fb
gattlib read write: Found bluetooth device '78:70:9B:A0:45:BD'
Read UUID completed: 68 69

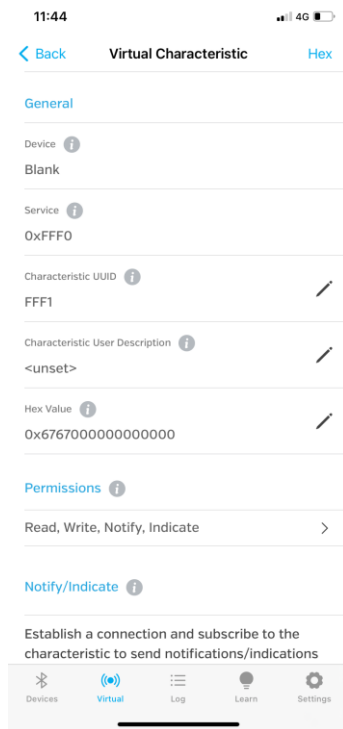
```

sudo ./examples/read_write/read_write 78:70:9B:A0:45:BD write 0000fff1-0000-1000-8000-00805f9b34fb 0x6767

```

dog50301@dog:~/gattlib/build$ sudo ./examples/read_write/read_write 78:70:9B:A0:45:BD write 0000fff1-0000-1000-8000-00805f9b34fb 0x6767
Value to write: 0x6767
gattlib read write: Found bluetooth device '78:70:9B:A0:45:BD'

```



```
sudo ./examples/notification/notification
```

```
→ sudo ./examples/notification/notification 78:70:9B:A0:45:BD 0000fff1-0000-1000-8000-00805f9b34fb
```

```
dog50301@dog:~/gattlib/build$ sudo ./examples/notification/notification 78:70:9B:A0:45:BD 0000fff1-0000-1000-8000-00805f9b34fb
gattlib notification: Found bluetooth device '78:70:9B:A0:45:BD' with RSSI:-33
gattlib notification: Wait for notification for 20 seconds...
Notification Handler: 68 69
```